nebio: CPU Measurement and Hardware Test

QB Nebulae V2 (RPi4) · branch **feature_newHardWareHandling** · measured 2026-10-16 on the device

1. Summary

nebio (libnebio.so) now builds and runs on the Nebulae. The hardware test passed: all inputs register, every LED works, and audio plays without underruns. With the SuperCollider bank running and nobody touching the module, the Pi is about **46 % busy** (all 4 cores = 100 %). Most of it is control overhead, not audio:

- **Python main loop of nebulae.py: 79 % of one core.** nebio delivers about 500 snapshots per second, so updateAll() runs about 500 times per second.
- **nebio input thread: 20 % of one core** (1 kHz scan of shift register, encoders and ADCs).
- **Hidden kernel work: about 17 % of the Pi** that top does not show under any process: about 8000 SPI interrupts/s (ADC reads), about 1000 I2C interrupts/s (LED writes) and about 9600 inter-core interrupts/s.
- **SuperCollider and JACK together: about 15 % of one core.**

The numbers are a baseline. They have to be compared with the original firmware under the same conditions before deciding what to optimise (section 7).

Status

Item	Result
libgpiod installed on the Pi	Done: libgpiod 2.3.1 (cmake, gcc, make, pkgconf, liblo 0.34 were already there)
nebio build on the Pi	Done: libnebio.so, 0 warnings (armv7h, gcc 15.2)
Hardware test firmware (hwtest.py)	Runs at boot, log on the USB stick
Inputs: knobs, CV, buttons, gates, encoders	All register (user test)
LEDs: 5 button LEDs, 4 RGB LEDs x 3 colours	All work (ledwalk.py, checked by eye)
Audio during the test	Plays, 0 underruns
Normal nebulae with nebio	Starts, SC bank plays
CPU baseline (this report)	Done for the SC bank; original firmware not measured yet

2. Setup

Item	Value
Device	Raspberry Pi 4, 4 cores, Arch Linux ARM **armv7h** (32 bit), kernel 5.15.43-2-rpi-ARCH
CPU clock	governor performance, 1200 MHz
Firmware	nebio firmware (libnebio.so built 2026-10-16 07:32), nebulae.py md5 e67c4c6f8289
Running	nebulae.py (python3), sclang, scsynth, jackd (-p128 -n3 -r48000, hw:sndrpiproto)
Instrument	SC bank, the instrument loaded at boot
Conditions	No user input, no USB activity, temperature 44-47 °C, never throttled
Window	300 s, one sample per second (273 intervals), 08:41-08:46 on the Pi clock
Tool	Code/scripts/cpustat.sh (sh + awk + /proc only, so it also runs on the original firmware)

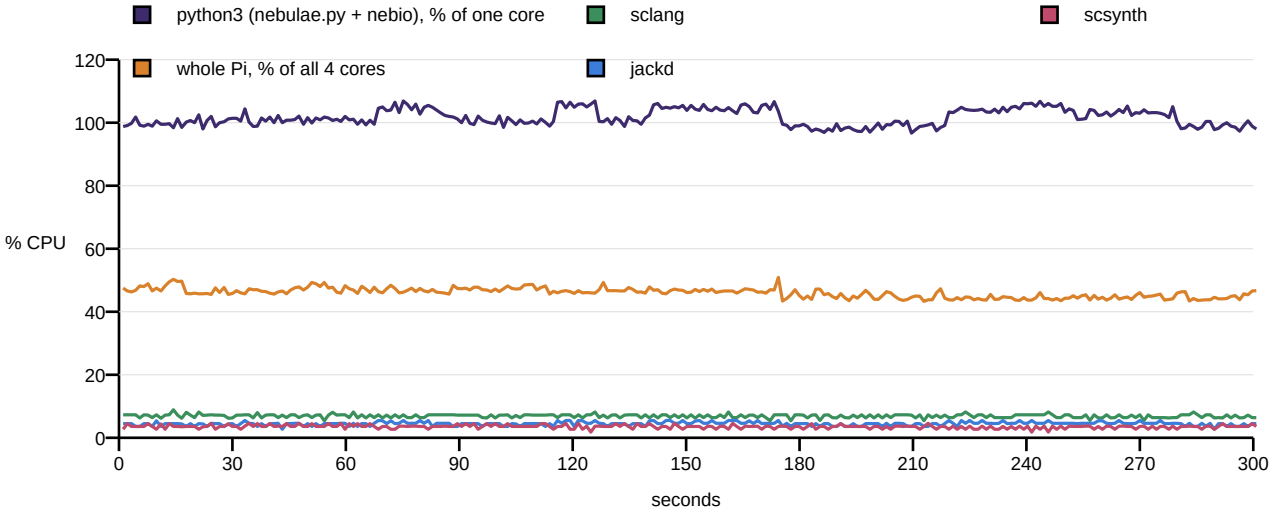
CPU values are given as **% of one core**, the same as top (100 % = one full core). "Share of Pi" is the same value divided by 4. The whole-Pi line uses /proc/stat, where all 4 cores = 100 %.

3. Results: SC bank, nebio firmware

Program	min %	mean %	p95 %	max %	share of Pi	max RSS
python3 (nebulae.py incl. nebio)	96.7	**101.7**	106.0	106.9	25.4 %	80 MB
sclang	5.4	6.9	7.3	8.9	1.7 %	29 MB

Program	min %	mean %	p95 %	max %	share of Pi	max RSS
scsynth	1.8	3.5	4.5	4.6	0.9 %	118 MB
jackd	2.7	4.5	5.5	5.5	1.1 %	127 MB
pd	-	-	-	-	not running	
csound	-	-	-	-	runs inside python3	
whole Pi (4 cores = 100 %)	43.3	**46.0**	48.4	50.9		
temperature (°C)	43.8	45.3	46.3	46.7		

CPU over the 5 minutes



The load is flat: there are no spikes, and nothing grows over time.

Inside the python3 process

nebio threads (and in the Csound bank also Csound) run inside nebulae.py, so the process is split per thread:

Thread	mean %	max %	What it is
main thread (tid 528)	**79.0**	84.2	Python main loop: updateAll(), ControlHandler, ui.py
tid 577	**20.2**	21.9	nebio input thread, 1 kHz scan
tid 559	1.6	2.7	nebio thread (LED writer or gate events)
tid 560	0.8	1.8	nebio thread
tid 578	0.0	0.9	nebio thread (idle)

The nebio threads are not named yet, so all show as python3. The input thread is identified by its steady 20 %, which matches the earlier hardware test.

4. Hidden kernel work: interrupts

The programs above add up to **29 %** of the Pi, but /proc/stat shows **46 %** busy (user 26.5 %, system 19.3 %, idle 54.0 %). The missing 17 % (about 0.7 of a core) belongs to no process. This kernel uses tick-based accounting without IRQ time accounting (CONFIG_HZ=100), so interrupt work that happens while a core is idle is counted as system time without any process. Interrupt rates over the 5 minutes:

Source	per second	Cause
IPI3 function call	9 582	Cores signalling each other (follows the SPI and thread wake-ups)
fe204000.spi	**7 996**	nebio ADC reads: 2 chips x 8 channels x 500 Hz = 8000, one interrupt per channel transfer
arch_timer	1 289	Kernel timer (1 kHz nebio sleep wake-ups)
fe804000.i2c	**986**	LED writes to the PCA9685, although nothing is being touched

Source	per second	Cause
DMA IRQ x 2	375 + 375	Audio stream, 48000 / 128 frames. Expected
IPI2 rescheduling	29	

New processes were also checked: only 2 were started in 10 s, so short-lived processes are not the cause.

5. Hardware test results

hwtest.py ran at boot instead of nebulae (firmware built with `create_fw_update.sh --hwtest`), with audio playing and the log written to the USB stick.

- **Inputs:** all knobs, both encoders, buttons and gates registered (118 digital events). CV 4 (pitch) and CV 1 (speed) were seen moving; the other CV inputs were not patched in that run.
- **nebio:** scan 1000 Hz, worst scan 1.0-2.4 ms, test process about 22 % of one core, whole Pi 1-20 %, 41-42 °C, not throttled.
- **Audio:** 12 files looped on plughw:sndrpiproto, **0 underruns**.
- **LEDs:** the PCA9685 registers were read while the test ran and matched the expected values. ledwalk.py then lit every LED and colour on its own; all worked. In hwtest.py the pitch/speed LEDs do not behave like nebulae (encoders only dim the _neg LEDs, knobs set the _pos colours), which looked like "not reacting".

6. Other findings

- **sudo** on the Pi asks for the root password, not alarm's.
- **pacman 7** needs `--disable-sandbox` on kernel 5.15 (no Landlock). The package database was stale (mirror 404); after `-Sy` glibc matched the repo, so libgpiod was installed on its own without a partial upgrade.
- **The Pi is 32 bit (armv7h)**, not aarch64 as the nebio step 1 report and the SC PKGBUILD assume.
- **The nebio build folder is owned by root** (built at boot by startup.sh), so manual builds must run as root.
- **USB stick dropped out during one boot** (sda "device offline or changed", I/O errors). Copying an .instr and a .pd file from the stick failed. Not related to nebio; the stick or the USB power should be checked.
- **Code/log is missing early in boot**, so the first neb_log.txt lines are lost. Harmless.

7. Possible improvements (not done)

To be decided after the comparison with the original firmware:

Change	Expected effect
Pace the Python main loop lower (for example 200 Hz instead of about 500 Hz)	Largest saving: the 79 % main thread scales with the loop rate. Knob smoothing filters work per loop and would need retuning.
ADC reads at 250 Hz instead of 500 Hz	Halves the SPI interrupts (8000/s) and part of the input thread
Write LEDs only when a value changes by more than a small threshold	Removes most of the 1000 I2C interrupts/s caused by knob noise
Name the nebio threads (pthread_setname_np)	top and cpustat.sh show nebio-input, nebio-leds, ... instead of python3

8. How to compare with the original firmware

Install the original firmware, load the **same SC instrument**, do not touch the module, copy the script to /tmp (the original firmware does not have it) and run:

```
sudo sh /tmp/cpustat.sh -d 300 -l "SC bank, original firmware"
# -> /tmp/cpustat_<date>.txt (report) and .csv (one row per second)
# /tmp is cleared at reboot: copy both files off before rebooting
```

Repeat for the Csound and Pure Data banks on both firmwares. In the Csound bank the python line includes Csound. Compare: python mean, whole-Pi mean, the system share and the SPI/I2C interrupt rates.

Files

File	Content
measurements/cpustat/nebio_sc_20261016_084140.txt	Raw report of this measurement
measurements/cpustat/nebio_sc_20261016_084140.csv	One row per second (chart data)
measurements/cpustat/make_report.py	Builds this PDF
Code/scripts/cpustat.sh	Measurement script
Code/nebio/tools/hwtest.py, ledwalk.py	Hardware test and LED walk